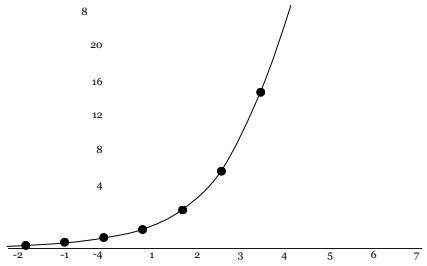
1.24 Graph Sketching: Exponentials and Logarithms

Chapter 14 · Original slide 4 · y={2}^{x}



x	-2	-1	0	1	2	3	4
y							

Note: Ensure that you can distinguish between a x^a (e.g. polynomial) term and an a^x exponential term. In the former the variable is in the base, and in the letter the variable is in the power. 2^x behaves very differently to x^2, both in its rate of growth (i.e. exponential terms grow much faster!) and how it differentiates.

Why are exponential functions important?

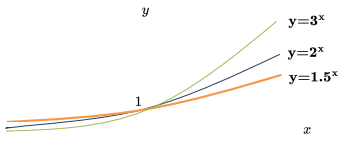
Each of the common graphs have a key property that makes them useful for modelling.

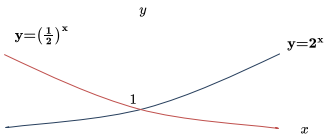
For reciprocal graphs $y = \frac{k}{x}$, as x doubles, y halves. This means we'd use it to represent variables which are inversely proportional.

Linear graphs are used when we're adding the same amount each time.

In contrast, exponential graphs are used when we're multiplying by the same amount each time. For example, we might use $1000(1.05^t)$ to model our savings with interest, where each year we have 1.05 times as much, i.e. with 5% added interest.

Chapter 14 · Original slide 5 · Contrasting exponential graphs





On the same axes sketch $y = 3^x, y = 2^x, y = 1.5^x$

On the same axes sketch $y = 2^x$ and $y = \left(\frac{1}{2}\right)^x$

Three important notes:

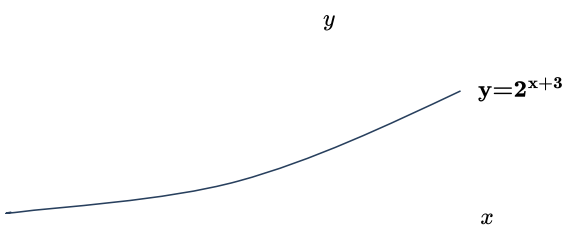
• $y = 2^x$ is said to be "exponential growing" whereas $y = \left(\frac{1}{2}\right)^x$ is said to be "exponentially decaying", because it's getting smaller (halving) each time x increases by 1.

• $y = \left(\frac{1}{2}\right)^x$ is a reflection of $y = 2^x$ in the line $x = 0$.
Proof:
If $f(x) = 2^x$, $f(-x) = 2^{-x} = \frac{1}{2^x} = \left(\frac{1}{2}\right)^x$

• $\left(\frac{1}{2}\right)^x$ would usually be written 2^{-x}.
You should therefore in general be able to recognise and sketch the graph $\mathbf{y=a^{-x}}$.

1.24 Graph Sketching: Exponentials and Logarithms

Chapter 14 · Original slide 6 · Graph Transformations



Sketch $y = 2^{x+3}$

The 'change' to 2^x is 'inside the function' (i.e. input x is replaced with $x + 3$).

Ensure you work out the new y-intercept.

Chapter 14 · Original slide 7 · Exercise 14A (Pages 313-314)

6 The graph of $y = ka^x$ passes through the points $(1, 6)$ and $(4, 48)$. Find the values of the constants k and a.

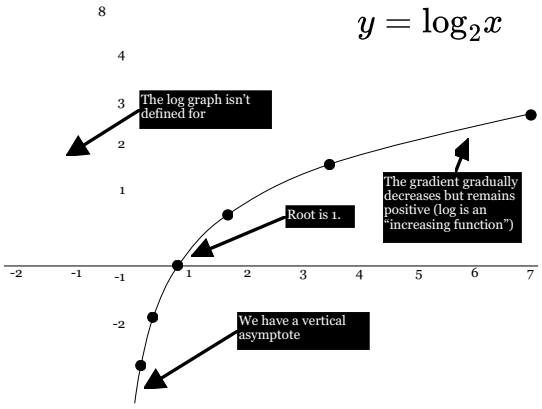
7 The graph of $y = pq^x$ passes through the points $(-3, 150)$ and $(2, 0.048)$.

a By drawing a sketch or otherwise, explain why $0 < q < 1$.

b Find the values of the constants p and q.

1.24 Graph Sketching: Exponentials and Logarithms

Chapter 14 · Original slide 27 · Exponentials and logarithms



x	0.25	0.5	1	2	4	8
y						